



Moriah College
MATHEMATICS DEPARTMENT

Mathematics

Year 12

Extension 2 Pre-Trial 2009

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General Instructions

- Reading time – 5 minutes
- Working time – 3 hours
- Write using blue or black pen
- Board-approved calculators may be used
- A table of standard integrals is provided.
- All necessary working should be shown in every question

Total marks (120)

- Attempt Questions 1–8
- All questions are of equal value
- Use a **SEPARATE** booklet for each question

STUDENT NUMBER: _____

Question 1

a) Find $\int_{-5}^1 \frac{dx}{x^2 + 4x + 13}$ **3**

b) Find $\int \frac{2dx}{x^2 - 4x}$ **3**

c) Use integration by parts to find the value of $\int_1^3 x^3 \log x dx$ **3**

d) Find $\int \frac{\sin x dx}{1 + \cos 2x}$ **3**

e) Using a suitable substitution, find $\int \frac{dx}{x^2 \sqrt{9 - x^2}}$ **3**

Question 2

a) Find all values of x and y such that $(x + iy)^2 = 4i$ **3**

b) Let $w = -2 + 2\sqrt{3}i$ **6**

i) Find $|w|$

ii) Show that $w^2 = 4\bar{w}$

iii) Prove that w is a root of the equation $z^3 - 64 = 0$.

iv) Find all roots of the equation $z^3 - 64 = 0$.

c) Draw neat sketches showing the main features of the following locii of z . **4**

i) $\frac{-\pi}{3} \leq \arg(z - 2) \leq \frac{\pi}{3}$

ii) $|z - 2 - 2i| \leq 2$

d) i) For z satisfying $|z - 2 - 2i| \leq 2$, find the greatest value of $\arg(z)$. **2**

ii) For z satisfying $|z - 2 - 2i| \leq 2$, find the greatest value of $\text{mod}(z)$

Question 3

a) Consider the ellipse $\frac{x^2}{4} + y^2 = 1$. 6

- i) Find the eccentricity, e , of this ellipse
- ii) Find the coordinates of the foci of the ellipse.
- iii) Use implicit differentiation to show that the derivative of the equation of the ellipse is given by $\frac{dy}{dx} = \frac{-x}{4y}$
- iv) Show that the gradient of the tangent at the endpoint of the latus rectum in the first quadrant is $-e$.

b) i) Find $\frac{(2-2i)(-\sqrt{3}+i)}{2i}$ in the form $a+ib$. 5

- ii) Express $2-2i$ in mod-arg form
- iii) Express $\frac{(2-2i)(-\sqrt{3}+i)}{2i}$ in mod-arg form
- iv) Hence find the exact value of $\sin 15^\circ$

c) Let $P(x) = 2x^3 + 3x^2 + 8x - 5$ 4

- i) Write down the only possible rational roots of the equation $P(x) = 0$
- ii) Hence solve completely, the equation $P(x) = 0$.

Question 4

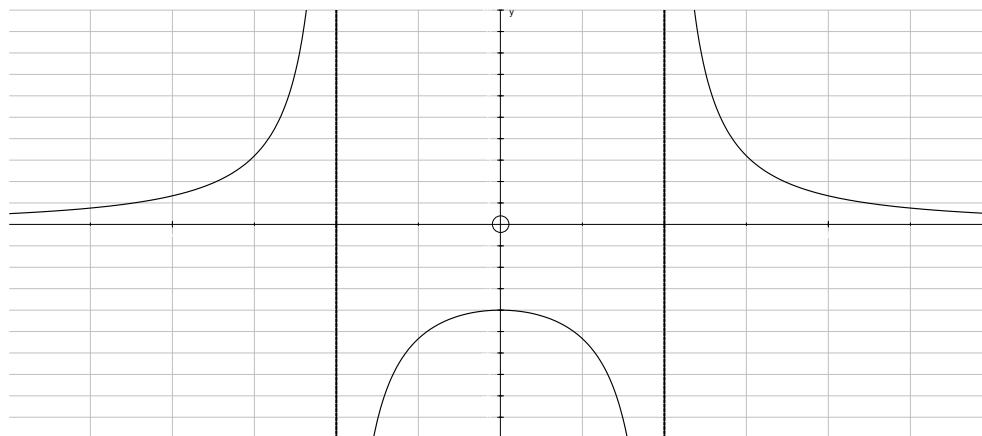
- a) 4
 Use the substitution $t = \tan \frac{x}{2}$ to show that $\int_0^{\frac{\pi}{2}} \frac{dx}{2 - \sin x} = \frac{2}{3\sqrt{3}} \pi$
- b) 4
 $P(cp, \frac{c}{p})$ lies on the rectangular hyperbola $xy = c^2$.
- i) Show that the equation of the normal at P is $p^3x - py - cp^4 + c = 0$
- ii) Hence, find the coordinates of the other point Q , where this normal cuts the hyperbola.
- c) 4
 A complex number z moves in the Argand plane so that $\operatorname{Re}(z) = 2|z - 3|$
- i) Explain why the locus defined by the above equation is an ellipse with one focus at $(3, 0)$.
- ii) Find the centre of the ellipse and the length of its major axis.
- d) 3
 Solve $x^4 + x^3 - 3x^2 - 5x - 2 = 0$ given that it has a root of multiplicity 3

Question 5

a)

8

The graph of $f(x) = \frac{1}{(x-4)(x+4)}$ is sketched below.



Draw neat sketches showing the main features of:

i) $y = \frac{1}{(4-x)(x+4)}$

ii) $y^2 = \frac{1}{(x-4)(x+4)}$

iii) $y = \frac{x}{(x-4)(x+4)}$

iv) $y = \tan^{-1}\left(\frac{1}{(x-4)(x+4)}\right)$

b)

7

i) Show that the equation $x^3 - 6x^2 + 9x - 5 = 0$ has only one real root, α

ii) Determine the two consecutive integers between which α lies.

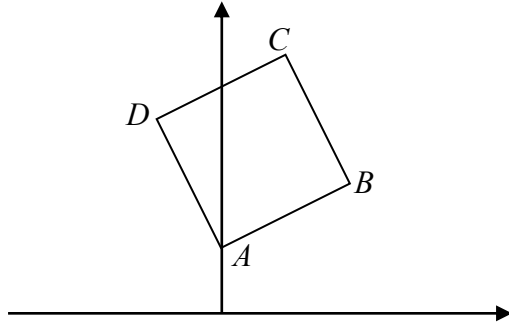
iii) Express the modulus of each of the complex roots in terms of α .

iv) Deduce that the value of this modulus lies between 1 and $\frac{\sqrt{5}}{2}$

Question 6

a) Find all the solutions of the equation $\tan 3x = \cot 2x$ 4

b) 4



The diagram shows the square $ABCD$ in the complex plane.
 A represents the complex number i and B represents the complex number z .

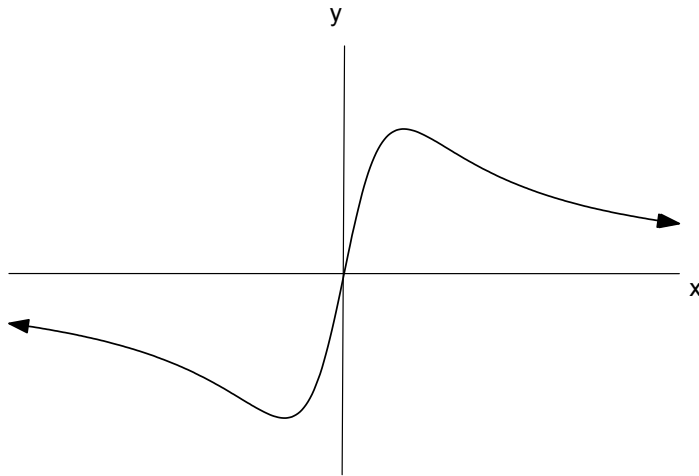
- i) Find the complex number represented by the vector $\overrightarrow{AD}$.
 - ii) Hence, or otherwise, find the complex number represented by the point C .
- c) 7
- i) Show that the graph of $y = e^x \sin x$ crosses the x -axis when $x = k\pi$ (k is an integer).
 - ii) Draw a sketch of $y = e^x \sin x$ in the domain $0 \leq x \leq 2\pi$.
 - iii) Let $I = \int_{(k-1)\pi}^{k\pi} e^x \sin x dx$ where k is an integer.
 Show that $2I = \frac{(-1)^{k-1} e^{k\pi} (e^\pi + 1)}{e^\pi}$
 - iv) Find the area bounded by $y = e^x \sin x$, the x -axis and the lines $x = 0$ and $x = 2\pi$

Question 7

- a) i) Show that the equation of the tangent to the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ ($a > b > 0$) at the point $P(a \sec \theta, b \tan \theta)$ is: **8**
- $$bx \sec \theta - ay \tan \theta = ab$$
- ii) If this tangent passes through a focus of the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$, show it must be parallel to $y = x$ or $y = -x$.
- iii) Show also that its point of contact with the hyperbola lies on a directrix of the ellipse.
- b) **7**
- i) If $I_n = \int_0^{\frac{\pi}{2}} x^n \cos x dx$, show that $I_n = \left(\frac{\pi}{2}\right)^n - n(n-1)I_{n-2}$
- ii) Hence find the value of I_4
- iii) Find the value of I_3

Question 8

- a) i) If w is a complex cube root of unity, show that $1 + w + w^2 = 0$ 4
- ii) Find the cubic equation whose roots are $1, 1 + w, 1 + w^2$.
- b) The graph of the function $f(x) = \frac{2x}{1+x^2}$ is shown below. 4



- i) Sketch the curve $y = |f(x)|$
- ii) Find the acute angle between the two branches of the curve $y = |f(x)|$, where they meet at the origin.

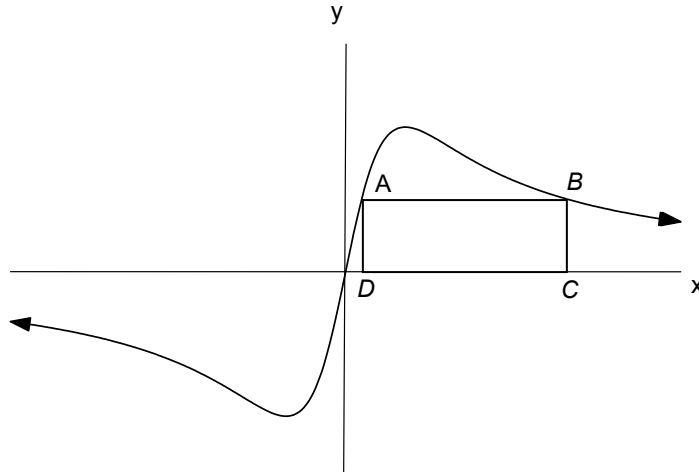
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c)

7

The graph of the function $f(x) = \frac{2x}{1+x^2}$ is again shown below.

In this diagram a rectangle $ABCD$ has been drawn. The inscribed rectangle has its base CD lying on the x -axis and its height given by the function.



- i) Show that $f(x) = f\left(\frac{1}{x}\right)$
- ii) The area below the curve and above the x -axis, in the first quadrant, is unlimited. Show that this is true by considering the definite integral $\int_0^a \frac{2x}{1+x^2} dx$.
- iii) Let C be the variable point $(x,0)$. Using part i) or otherwise, find an expression for the area of the inscribed rectangle $ABCD$ in terms of x .
- iv) Hence, find the limit of the area of the inscribed rectangle

Question one

$$\begin{aligned} \text{a) } \int_{-5}^1 \frac{dx}{x^2 + 4x + 13} &= \int_{-5}^1 \frac{dx}{(x+2)^2 + 9} \\ &= \left[\frac{1}{3} \tan^{-1} \frac{x+2}{3} \right]_{-5}^1 \\ &= \frac{1}{3} [\tan^{-1} 1 - \tan^{-1} -1] = \frac{\pi}{6} \end{aligned}$$

$$\begin{aligned} \text{b) } \text{Let } \frac{2}{x(x-4)} &= \frac{A}{x-4} + \frac{B}{x} \\ Ax + B(x-4) &= 2 \\ x=0 \Rightarrow B &= \frac{-1}{2}, x=4 \Rightarrow A = \frac{1}{2} \end{aligned}$$

So, integral becomes

$$\frac{1}{2} \int \frac{1}{x-4} - \frac{1}{x} dx = \frac{1}{2} \ln \left| \frac{x-4}{x} \right| + C$$

$$\text{c) } \int_1^3 x^3 \log x \, dx$$

$$u' = x^3 \Rightarrow u = \frac{x^4}{4}$$

$$v = \log x \Rightarrow v' = \frac{1}{x}$$

$$\text{So, } \int_1^3 x^3 \log x \, dx = \left[\frac{x^4 \log x}{4} \right]_1^3 - \int_1^3 \frac{x^3}{4} dx$$

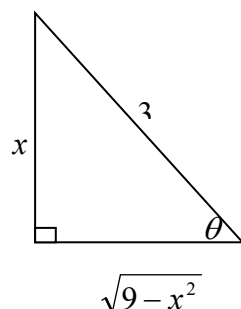
$$= \frac{81 \ln 3}{4} - \left[\frac{x^4}{16} \right]_1^3 = \frac{81 \ln 3}{4} - 5$$

$$\text{d) } = \int \frac{\sin x \, dx}{2 \cos^2 x} = \frac{1}{2} \int \sin x \cos^{-2} x \, dx = -\frac{1}{2} \frac{(\cos x)^{-1}}{-1} = \frac{1}{2} \sec x + C$$

$$\text{e) Let } x = 3 \sin \theta, \frac{dx}{d\theta} = 3 \cos \theta. \text{ Integral becomes;}$$

$$\int \frac{3 \cos \theta \, d\theta}{9 \sin^2 \theta \sqrt{9 - 9 \cos^2 \theta}} = \int \frac{d\theta}{9 \sin^2 \theta} = \frac{1}{9} \int \operatorname{cosec}^2 \theta \, d\theta$$

$$= \frac{-\cot \theta}{9} = \frac{-\sqrt{9-x^2}}{9x} + C$$

**Question two**

$$\begin{aligned} \text{a) } x^2 - y^2 + 2ixy &= 4i \\ \Rightarrow x^2 - y^2 &= 0 \text{ \& } xy = 2 \\ \therefore x &= \pm\sqrt{2}, y = \pm\sqrt{2} \end{aligned}$$

$$\text{b)i) } |w| = 4$$

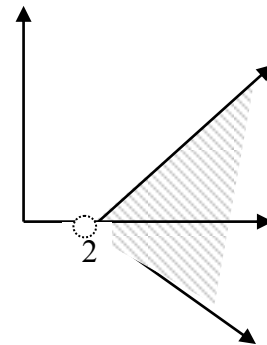
$$\begin{aligned} \text{ii) LHS} &= w^2 = -8 - 8\sqrt{3}i \\ \text{RHS} &= 4\bar{w} = 4(-2 - 2\sqrt{3}i) = -8 - 8\sqrt{3}i \\ &= \text{LHS} \end{aligned}$$

$$\text{iii) } \arg w = \frac{2\pi}{3}$$

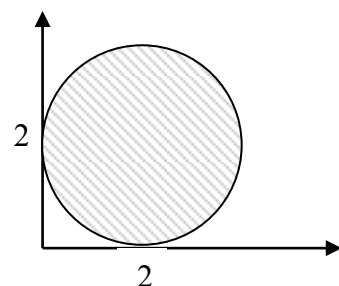
$$\text{So } w = 4\operatorname{cis}\left(\frac{2\pi}{3}\right) \text{ and } w^3 = 64\operatorname{cis}(2\pi) = 64.$$

$$\text{iv) Since coefficients are real, roots are } w, \bar{w}, \text{ and } 4, \text{ i.e. } -2 \pm 2\sqrt{3}i, 4.$$

c) i)



ii)



c)

$$\text{i) Max value } \arg z = \frac{\pi}{2}$$

$$\text{ii) greatest value of } |z| = 2\sqrt{2} + 2$$

Question three

a)

$$\text{i) } e^2 = 1 - \frac{b^2}{a^2} = 1 - \frac{1}{4} \Rightarrow e = \frac{\sqrt{3}}{2}$$

$$\text{ii) Foci are } (\pm ae, 0) = (\pm\sqrt{3}, 0)$$

$$\text{iii) } \frac{x^2}{4} + y^2 = 1 \text{ - differentiate both sides:}$$

$$\frac{2x}{4} + 2y \frac{dy}{dx} = 0$$

$$2y \frac{dy}{dx} = \frac{-x}{2} \Rightarrow \frac{dy}{dx} = \frac{-x}{4y}$$

iv) Find end point of the latus rectum;

$$x = \sqrt{3} \Rightarrow y = \sqrt{1 - \frac{3}{4}} = \frac{1}{2}$$

$$\therefore \frac{dy}{dx} = \frac{-x}{4y} = \frac{-\sqrt{3}}{4 \times \frac{1}{2}} = \frac{-\sqrt{3}}{2} = -e$$

b)i)

$$= \frac{-2\sqrt{3} + 2i + 2i\sqrt{3} + 2}{2i} \times \frac{2i}{2i}$$

$$= \frac{2(1 - \sqrt{3}) + 2i(1 + \sqrt{3})}{-4} \times 2i$$

$$= \frac{-(1 + \sqrt{3}) + i(1 - \sqrt{3})}{-1} = \sqrt{3} + 1 + i(\sqrt{3} - 1)$$

$$\text{ii) } 2 - 2i = 2\sqrt{2} \text{cis}\left(\frac{-\pi}{4}\right)$$

$$\text{iii) } = \frac{2\sqrt{2}\left(\frac{-\pi}{4}\right) 2\text{cis}\left(\frac{5\pi}{6}\right)}{2\text{cis}\left(\frac{\pi}{2}\right)} = 2\sqrt{2} \text{cis}\left(\frac{\pi}{12}\right)$$

$$\text{iv) } \text{Im}(\sqrt{3} + 1 + i(\sqrt{3} - 1)) = 2\sqrt{2} \sin 15^\circ$$

$$\therefore \sin 15^\circ = \frac{\sqrt{3} - 1}{2\sqrt{2}}$$

c) i) the only possible rational roots are:

$$\pm \frac{1}{2}, \pm \frac{5}{2}, \pm 1, \pm 5$$

$$\text{ii) } P\left(\frac{1}{2}\right) = 0$$

$$\text{So, } P(x) = (2x - 1)(x^2 + 2x + 5)$$

$$\text{And the roots are } \frac{1}{2}, -1 \pm 2i$$

Question four

a) Let $t = \tan \frac{x}{2}$.

When $x = 0$, $t = 0$; $x = \frac{\pi}{2}$, $t = 1$

$$\frac{dt}{dx} = \frac{1}{2} \sec^2 \frac{x}{2} = \frac{1}{2} (\tan^2 \frac{x}{2} + 1) = \frac{1}{2} (t^2 + 1) \Rightarrow$$

$$dx = \frac{2dt}{t^2 + 1}$$

$\sin x = \frac{2t}{1+t^2}$, so integral becomes:

$$\int_0^1 \frac{2dt}{(1+t^2)(2 - \frac{2t}{1+t^2})} = \int_0^1 \frac{dt}{1+t^2 - t}$$

Complete the square; integral becomes:

$$\int_0^1 \frac{dt}{(t - \frac{1}{2})^2 + \frac{3}{4}} = \left(\frac{2}{\sqrt{3}} \tan^{-1} \frac{(2t-1)}{\sqrt{3}} \right)_0^1$$

$$= \frac{2}{\sqrt{3}} \left(\frac{\pi}{6} - \frac{-\pi}{6} \right) = \frac{2\pi}{3\sqrt{3}}$$

b)

i) $y = c^2 x^{-1} \Rightarrow y' = -c^2 x^{-2}$

At P, m tangent is $\frac{-1}{p^2}$, m normal is p^2 .

Equation of normal:

$$(y - \frac{c}{p}) = p^2(x - cp)$$

$$py - c = p^3(x - cp)$$

$$py - c = p^3x - cp^4$$

$$p^3x - py - cp^4 + c = 0.$$

ii) Let Q be $(cq, \frac{c}{q})$, $m_{QP} = p^2$.

$$m_{QP} = \frac{-1}{pq} = p^2 \Rightarrow q = \frac{-1}{p^3}. \text{ So } Q \text{ is:}$$

$$\left(\frac{-c}{p^3}, -cp^3 \right).$$

Question four

c)i) $|z - 3|$ - distance of z from $(3,0)$

$$= \frac{1}{2} \text{ its distance from the } y \text{ axis.}$$

$\Rightarrow$ ellipse with focus $(3,0)$ and directrix the y axis.

ii)

It follows from i) that the ellipse has centre $(4,0)$, major axis has length 4 units.

d) If multiple root is α ,

$$P(\alpha) = P'(\alpha) = P''(\alpha) = 0.$$

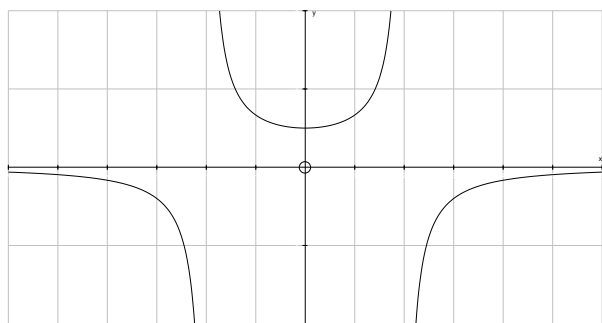
So, $12x^2 + 6x - 6 = 0$ has solution α .

$$\alpha = -1, \frac{1}{2}.$$

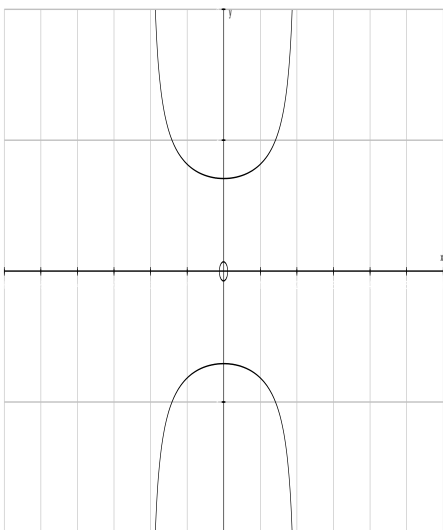
$\Rightarrow$ roots are $-1, -1, -1$, and 2 .

Question five

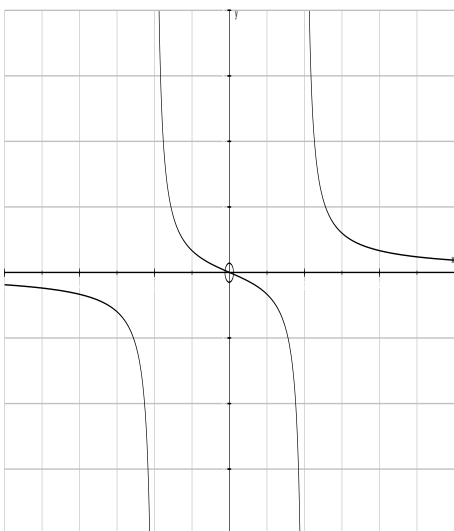
a) i)



ii)



iii)



b) i) Consider the graph of the function,

$$f(x) = x^3 - 6x^2 + 9x - 5 = 0.$$

Find the turning points. They are: (1,-1) (max) and (3,-5) (min). Since they are both below the x axis, the graph of $y = f(x)$ only crosses the x axis once, and hence the equation only has one real root, α .

ii) $f(4) = -1 < 0$ and $f(5) = 15 > 0$, so $4 < \alpha < 5$.

iii) Let the complex roots be z & $\bar{z}$.

Product of the roots:

$$\alpha z \bar{z} = 5$$

$$\therefore \alpha |z|^2 = 5$$

$$\therefore |z| = \sqrt{\frac{5}{\alpha}}$$

$$4 < \alpha < 5$$

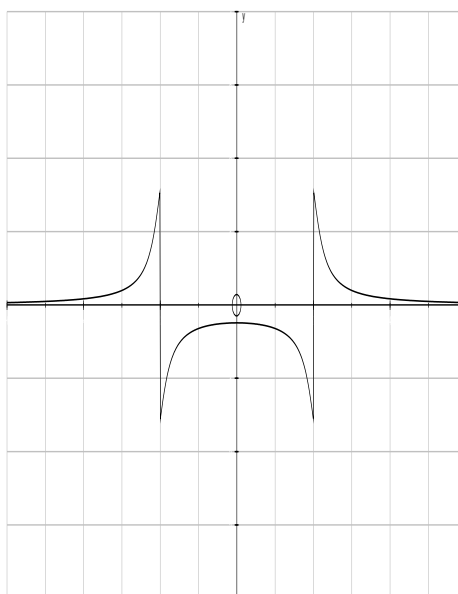
$$\therefore \frac{1}{5} < \frac{1}{\alpha} < \frac{1}{4}$$

$$1 < \frac{5}{\alpha} < \frac{5}{4}$$

$$1 < \sqrt{\frac{5}{\alpha}} < \frac{\sqrt{5}}{2}$$

$$1 < |z| < \frac{\sqrt{5}}{2}$$

a) iv)



Question six a) $\tan 3x = \cot 2x$

$$\tan 3x = \tan\left(\frac{\pi}{2} - 2x\right)$$

$$3x = k\pi + \frac{\pi}{2} - 2x$$

$$5x = k\pi + \frac{\pi}{2}$$

$$x = \frac{k\pi}{5} + \frac{\pi}{10} = \frac{\pi}{10}(2k+1) \quad (\cos 3x \neq 0)$$

b) i) $\overrightarrow{AB} = z - i$
 $\Rightarrow \overrightarrow{AD} = i(z - i) = 1 + iz$

ii) Let this number be u .

$$u - i = z - i + 1 + iz \Rightarrow u = 1 + z + iz$$

c) i) $x = k\pi, y = e^{k\pi} \sin k\pi = e^{k\pi} \times 0 = 0$.



iii) $I = \int_{(k-1)\pi}^{k\pi} e^x \sin x \, dx$ - integrate by parts.

$$u = e^x, u' = e^x \text{ \& } v' = \sin x, v = -\cos x$$

$$I = [-e^x \cos x] + \int_{(k-1)\pi}^{k\pi} e^x \cos x \, dx, \text{ Use parts again:}$$

$$u = e^x \quad u' = e^x$$

$$v' = \cos x \quad v = \sin x$$

$$I = [-e^x \cos x] + [e^x \sin x] - \int_{(k-1)\pi}^{k\pi} e^x \sin x \, dx$$

$$I = [-e^x \cos x] + [e^x \sin x] - I \dots \dots \dots \text{so,}$$

$$2I = [e^x (\sin x - \cos x)]_{(k-1)\pi}^{k\pi}$$

$$e^{k\pi} (\sin k\pi - \cos k\pi) - e^{(k-1)\pi} (\sin(k-1)\pi - \cos(k-1)\pi)$$

$$= e^{k\pi} (-\cos(k\pi) + e^{(k-1)\pi} \cos(k-1)\pi)$$

k even: =

$$e^{k\pi} (-1) + e^{(k-1)\pi} (-1) = e^{(k-1)\pi} (-1)(e^\pi + 1)$$

k odd: =

$$e^{(k-1)\pi} (e^\pi + 1) \Rightarrow 2I = \frac{(-1)^{k-1} e^{k\pi} (e^\pi + 1)}{e^\pi}$$

Question six (cont'd)

$$\text{Area} = \int_0^\pi e^x \sin x \, dx + \left| \int_\pi^{2\pi} e^x \sin x \, dx \right|$$

$$= \frac{e^\pi (e^\pi + 1)}{2e^\pi} + \frac{e^{2\pi} (e^\pi + 1)}{2e^\pi} = \frac{(e^\pi + 1)^2}{2} u^2$$

Question seven

i) Find $\frac{dy}{dx}$ (implicitly):

$$\frac{2x}{a^2} - \frac{2y}{b^2} \frac{dy}{dx} = 0 \Rightarrow \frac{dy}{dx} = \frac{b^2 x}{a^2 y} = \frac{b^2 a \sec \theta}{a^2 b \tan \theta} = \frac{b \sec \theta}{a \tan \theta} \text{ at } P.$$

$\therefore$ Equation tangent is:

$$y - b \tan \theta = \frac{b \sec \theta}{a \tan \theta} (x - a \sec \theta)$$

$$a \tan \theta y - ab \tan^2 \theta = b \sec \theta x - ab \sec^2 \theta$$

$$b \sec \theta x - a \tan \theta y = ab(\sec^2 \theta - \tan^2 \theta) = ab$$

ii) Focus of the ellipse: $\left(a \sqrt{\frac{a^2 - b^2}{a^2}}, 0 \right) = \left(\pm \sqrt{a^2 - b^2}, 0 \right)$ lies

on the above tangent. So, substituting:

$$\pm b \sec \theta \sqrt{a^2 - b^2} = ab$$

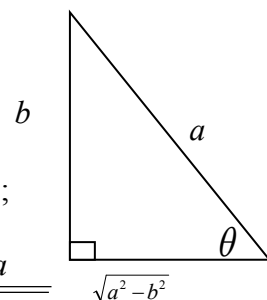
$$\sec \theta = \frac{\pm a}{\sqrt{a^2 - b^2}}$$

$$\tan \theta = \frac{b}{\sqrt{a^2 - b^2}}$$

$\therefore$ gradient of the tangent is;

$$\frac{b \sec \theta}{a \tan \theta} = \frac{\pm ab}{\sqrt{a^2 - b^2}} \div \frac{ba}{\sqrt{a^2 - b^2}} = \pm 1$$

And the result follows.



iii) Point of contact with the hyperbola is:

$$(a \sec \theta, b \tan \theta) = \left(\frac{a^2}{\sqrt{a^2 - b^2}}, b \tan \theta \right)$$

$$= \left(a \div \frac{\sqrt{a^2 - b^2}}{a}, \dots \dots \dots \right)$$

$$= \left(\frac{a}{e}, \dots \dots \right)$$

$\therefore$ lies on a directrix of the ellipse.

Question seven (cont'd)

i) Using parts:

$$u = x^n, u' = nx^{n-1}; v' = \cos x, v = \sin x$$

$$I_n = \left[x^n \sin x \right]_0^{\frac{\pi}{2}} - \int_0^{\frac{\pi}{2}} nx^{n-1} \sin x dx$$

$$= \left(\frac{\pi}{2} \right)^n - \int_0^{\frac{\pi}{2}} nx^{n-1} \sin x dx \rightarrow \text{use parts:}$$

$$u = x^{n-1}, u' = (n-1)x^{n-2}; v' = \sin x, v = -\cos x, \text{ so}$$

$$I_n = \left(\frac{\pi}{2} \right)^n - n \left(\left[-x^{n-1} \cos x \right]_0^{\frac{\pi}{2}} + (n-1) \int_0^{\frac{\pi}{2}} x^{n-2} \cos x dx \right)$$

$$= \left(\frac{\pi}{2} \right)^n - n(0) - n(n-1)I_{n-2}$$

As required.

$$I_4 = \left(\frac{\pi}{2} \right)^4 - 12I_2$$

$$\text{ii) } = \left(\frac{\pi}{2} \right)^4 - 12 \left(\frac{\pi}{2} \right)^2 + 12 \times 2I_0$$

$$I_0 = \int_0^{\frac{\pi}{2}} \cos x dx = 1 \therefore I_4 = \left(\frac{\pi}{2} \right)^4 - 12 \left(\frac{\pi}{2} \right)^2 + 24$$

$$\text{iii) } I_3 = \left(\frac{\pi}{2} \right)^3 - 6I_1 = \left(\frac{\pi}{2} \right)^3 - 6 \int_0^{\frac{\pi}{2}} x \cos x dx$$

Use parts: $u = x, u' = 1; v' = \cos x, v = \sin x$

$$\left[x \sin x \right]_0^{\frac{\pi}{2}} - \int_0^{\frac{\pi}{2}} \sin x dx = \frac{\pi}{2} - 1$$

$$\therefore I_3 = \left(\frac{\pi}{2} \right)^3 - 6 \frac{\pi}{2} + 6$$

Question eight

$$\text{a) i) } w^3 = 1 \Rightarrow w^3 - 1 = 0 \Rightarrow (w-1)(w^2 + w + 1) = 0$$

$$w \neq 1, \therefore w^2 + w + 1 = 0$$

$$\text{ii) Sum of roots: } 2 + w^2 + w + 1 = 2$$

In pairs

$$= 1 + w + 1 + w^2 + 1 + w + w^2 + w^3 = 1 + w^3 = 2$$

$$\text{Product of roots; } = (1+w)(1+w^2) = 1$$

 $\therefore$ Equation is:

$$x^3 - 2x^2 + 2x - 1 = 0$$

b)

Question eight (cont'd)

$$\text{b) ii) } f'(x) = \frac{2-2x^2}{(1+x^2)^2}$$

$$f'(0) = 2 \Rightarrow \tan \theta = \left| \frac{2-2}{1+2 \times -2} \right| = \frac{4}{3} \Rightarrow \theta = 53^\circ 8'$$

$$\text{c) i) } f\left(\frac{1}{x}\right) = \frac{\frac{2}{x}}{1 + \frac{1}{x^2}} \times \frac{x^2}{x^2} = \frac{2x}{1+x^2} = f(x)$$

ii)

$$\int_0^a \frac{2x}{1+x^2} dx = \left[\ln(1+x^2) \right]_0^a = \ln(1+a^2) \rightarrow \infty \text{ as } a \leftarrow \infty$$

$$\text{(iii) } D = \left(\frac{1}{x}, 0 \right)$$

$$\text{Area}(x) = \left(x - \frac{1}{x} \right) \left(\frac{2x}{1+x^2} \right) = \frac{2(x^2-1)}{1+x^2}$$

$$\text{iv) As } x \rightarrow \infty, A(x) \rightarrow 2$$